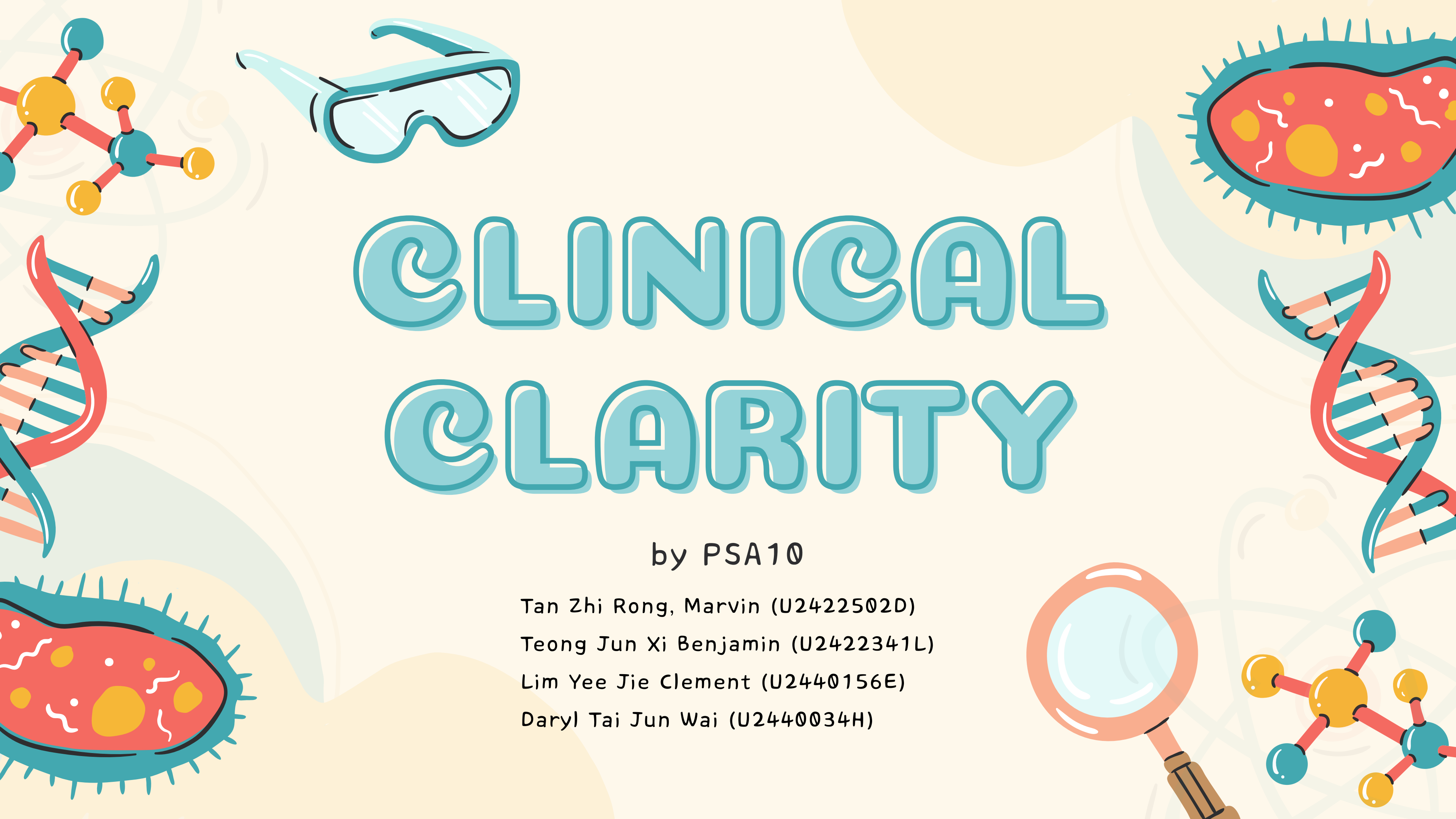




CLINICAL CLARITY

by PSA10

Tan Zhi Rong, Marvin (U2422502D)

Teong Jun Xi Benjamin (U2422341L)

Lim Yee Jie Clement (U2440156E)

Daryl Tai Jun Wai (U2440034H)

Clinical Dilemma

1 in 10

women have PCOS

~116M globally

1 in 9

women have Endometriosis

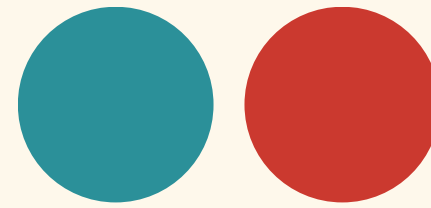
~190M globally

7-10 Years

average delay to diagnosis

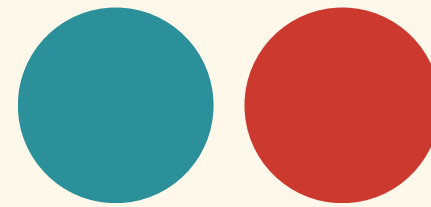
for endometriosis

Why is differentiation so hard?



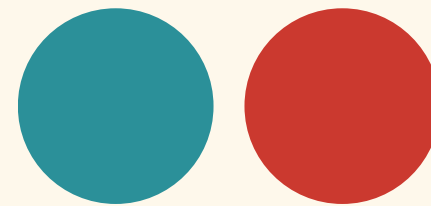
Pelvic Pain

shared by both conditions



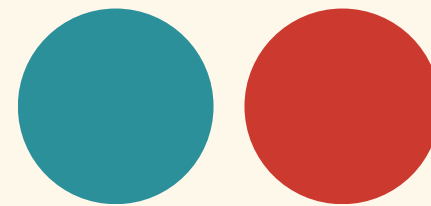
Irregular Menstrual Cycle

shared by both conditions



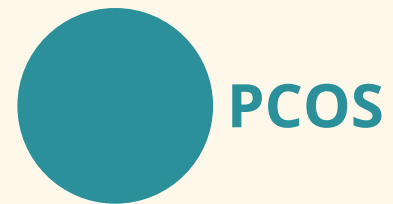
Hormonal Imbalances

shared by both conditions

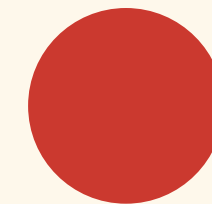


Infertility Concerns

shared by both conditions



PCOS



Endometriosis

Solution

3-Layer Diagnostic Pathway

PCOS Screening

1

- Random Forest on 41 clinical features (hormones, ultrasound, symptoms, demographics)
- Aligned with Rotterdam Criteria

PCOS Likely (>60%)

Endometriosis Check

2

Second Random Forest on 6 endo-specific features:

- Chronic Pain
- BMI
- Age
- Menstrual Irregularity
- Hormone Abnormality
- Infertility

Endo Possible (>60%)

Rule-Out Workup

3

- TSH and Prolactin threshold checks
- Flags Hypothyroidism/Hyperprolactinemia
- Refer to Reproductive Endocrinology

Specialist Referral

Datasets: PCOS & Endometriosis

PCOS Dataset

Main Clinical Dataset

Patients	541
Features	41 Clinical
Positive (PCOS)	177 (32.7%)
Negative	364
Age range	20 - 48 yrs
BMI range	12.4 - 38.9
Source	Single center (India)

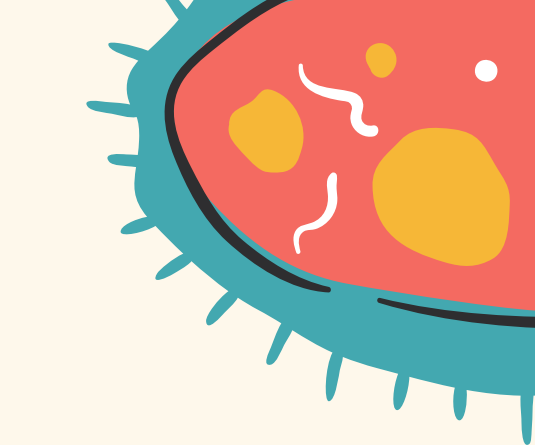
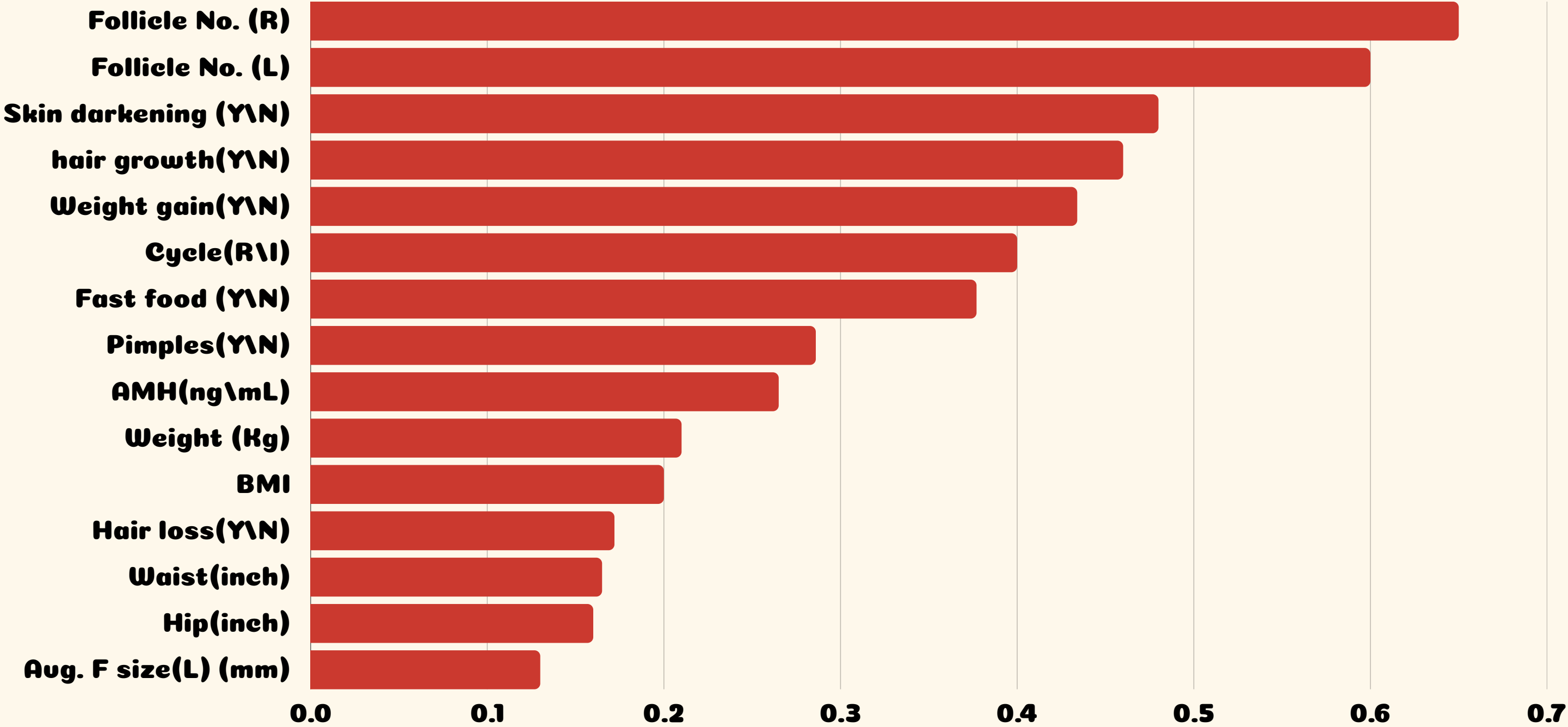
Endometriosis Dataset

Structured Supplementary Dataset

Patients	10,000
Features	6 Clinical
Positive (PCOS)	4,079 (40.8%)
Negative	5,921 (59.2%)
Age range	15 -55 Yrs
Mean Pain Level	5.03 / 10
Dataset type	Structured Synthetic

Exploratory Data Analysis

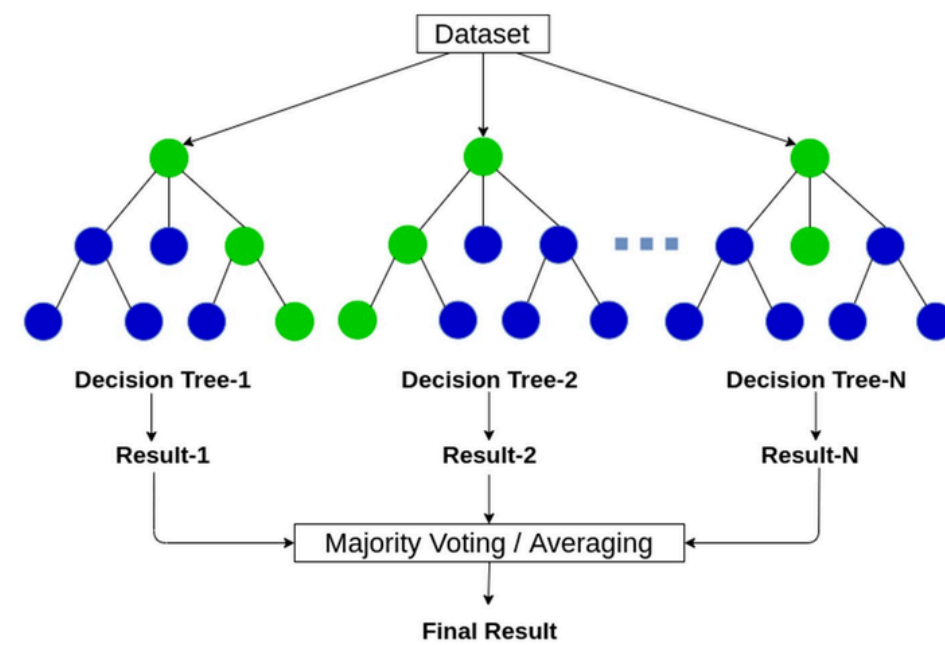
Top 15 Clinical Predictors of PCOS(Pearson Coeff.)



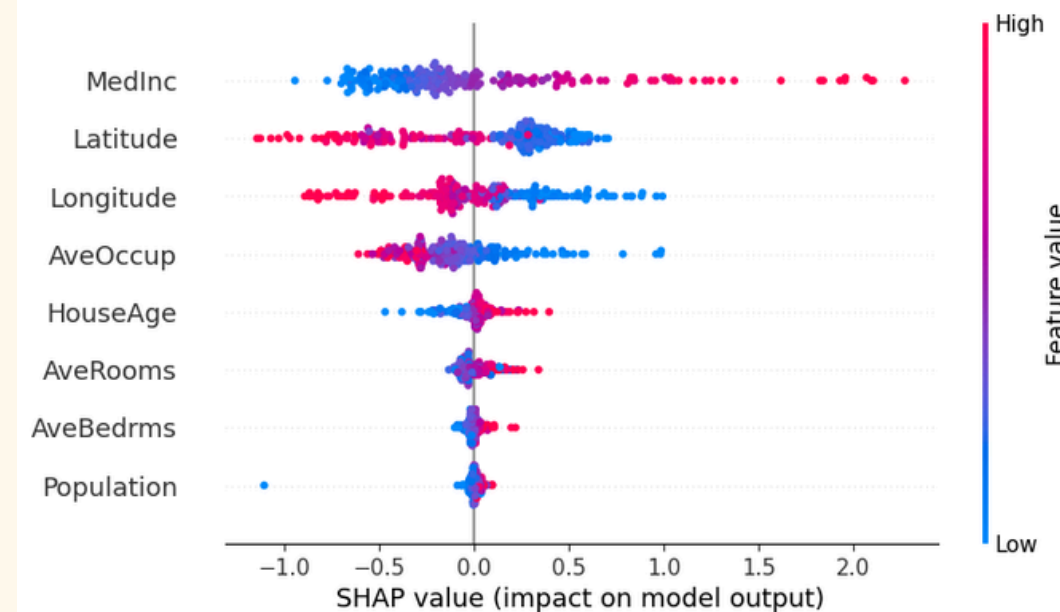
Machine Learning Architecture

The Predictive Engine

Random Forest



Clinical Transparency



Random Forest

01.

Layer 1 - PCOS Classifier

More than 60 → PCOS LIKELY

If not check for Endometriosis

02.

Layer 2 - Endometriosis Classifier

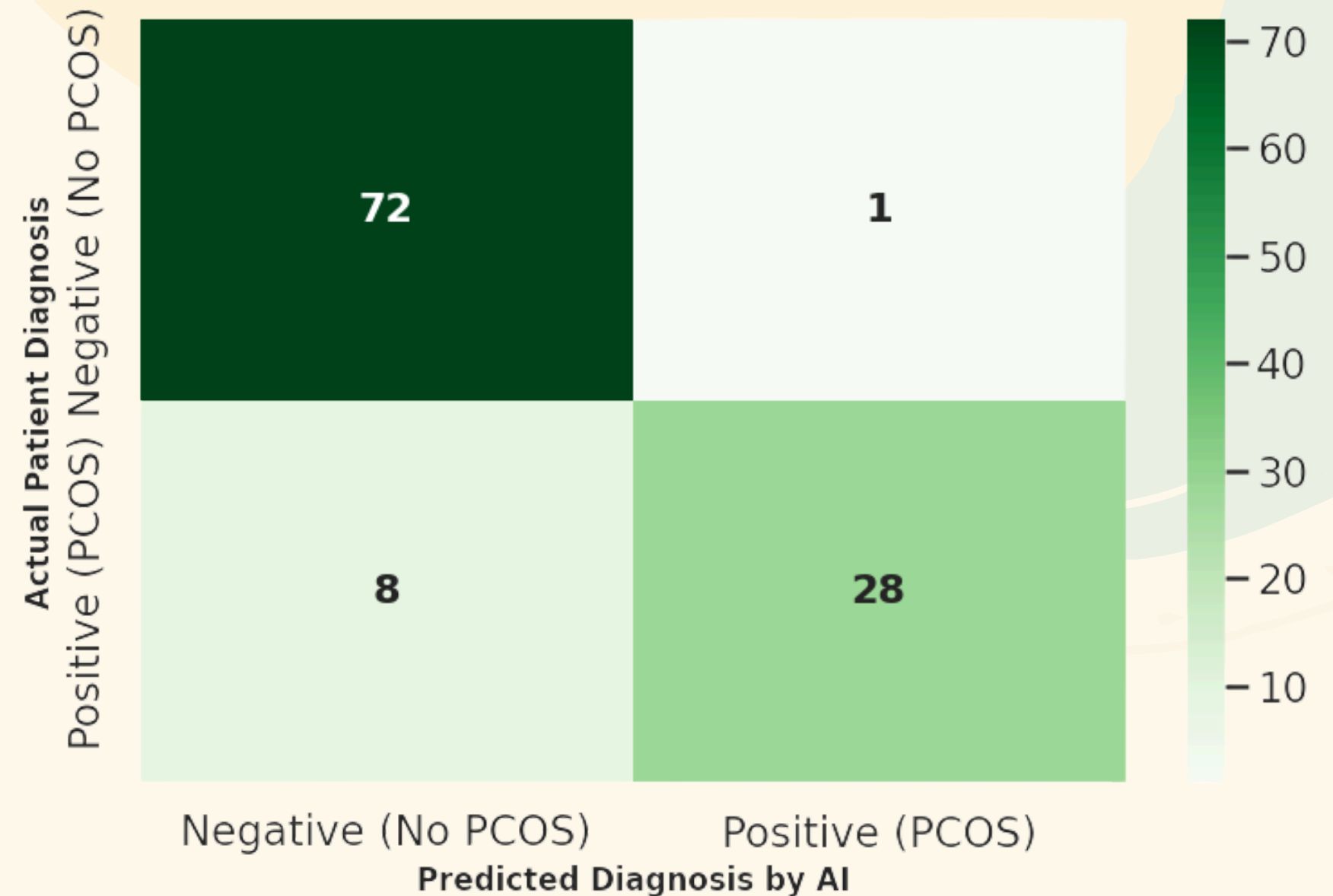
More than 60% → Endo Possible

03.

Layer 3 - Clinical Rule-Out

If none → Specialist referral

PCOS Diagnostic Confusion Matrix
(Tested on Unseen Patient Data)



Our model has a 91.74% accuracy
100 out of 109 patients predicted
correctly

Explainable AI (SHAP) – Improve Model Accuracy

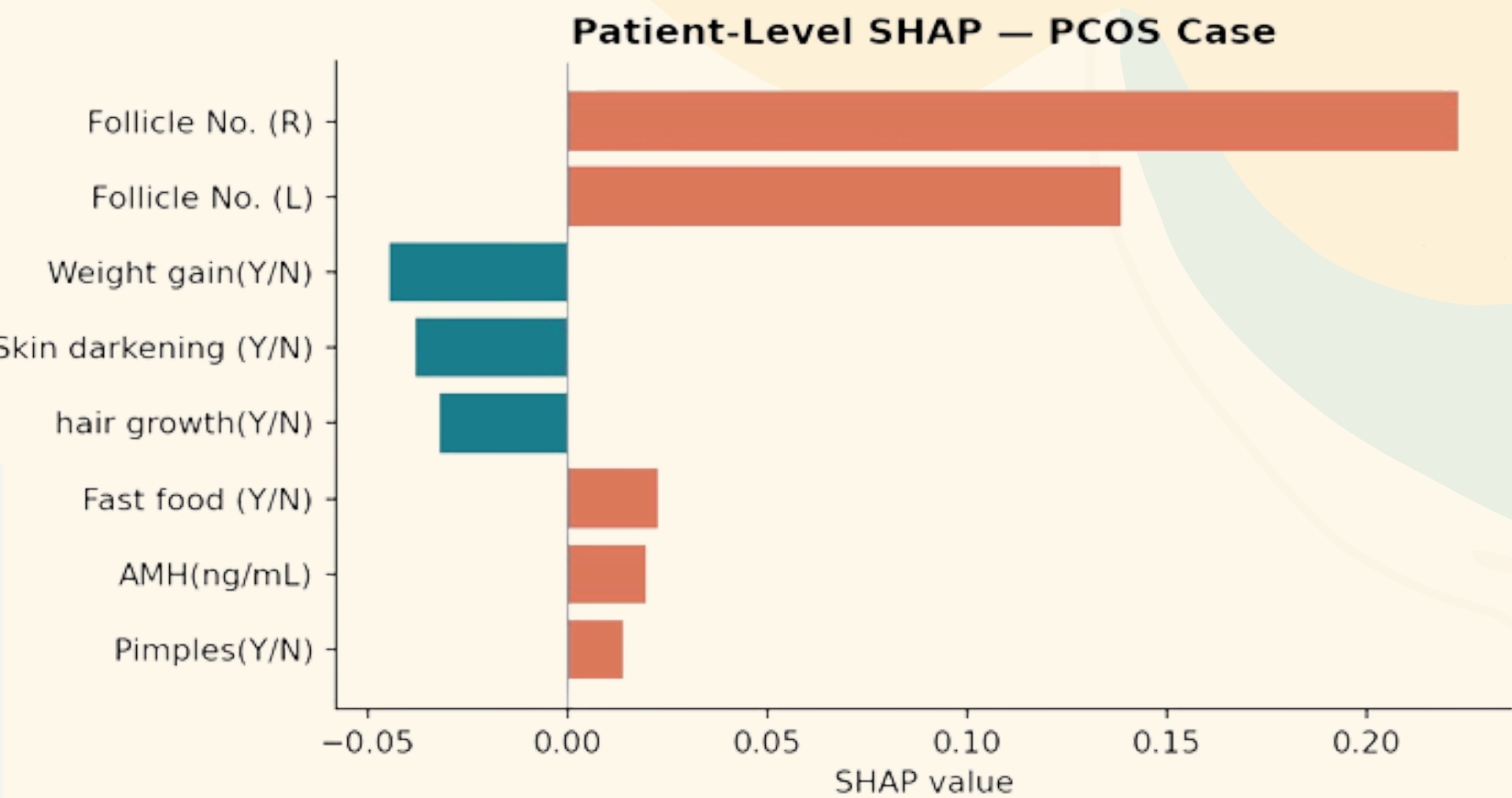
Why SHAP?

SHAP (SHapley Additive exPlanations) assigns each feature a contribution score for a specific prediction. For clinical tools, this is critical, it lets clinicians see WHY the model flagged PCOS, not just that it did.

Top 10 SHAP Features (PCOS model, mean |SHAP|)

Follicle No. (R)	0.131
Follicle No. (L)	0.070
Weight gain	0.053
Skin darkening	0.052
Hair growth	0.043
Cycle (R/I)	0.027
AMH (ng/mL)	0.025
Fast food	0.024
Pimples	0.016

Patient-Level SHAP — Sample PCOS Case



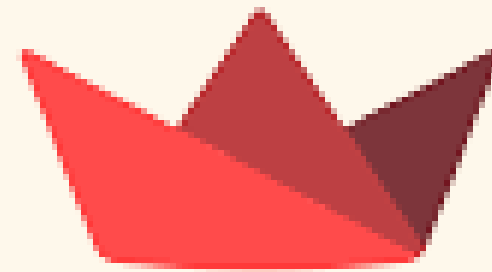
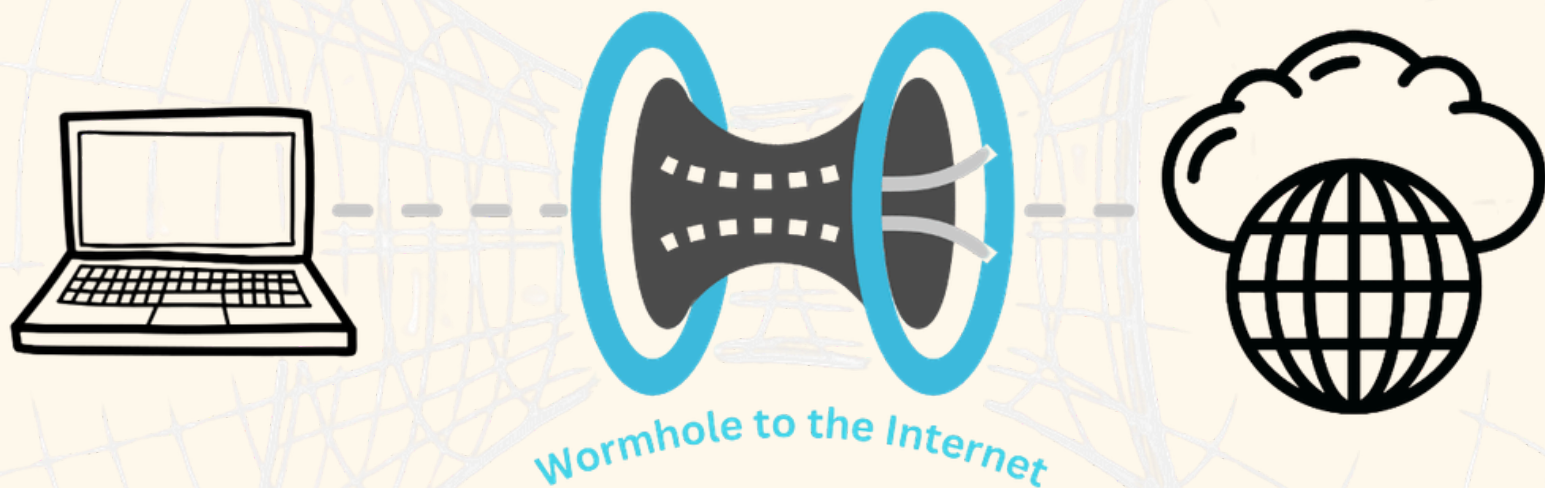
Interpretation:

- Red bars = pushes prediction toward PCOS
- Blue bars = pushes prediction away from PCOS
- Length = magnitude of influence

Software Used for UI

Local Tunnel

Test & share local services globally



Streamlit

DEMO: PCOS High Probability (Layer 1)

Key Parameter: Follicle Count(L\|R)

Entered values look consistent with the training data.

> See full feature vector sent to the model (including medians)

Clinical summary

Layer 1 — PCOS probability: 91.0%

Layer 1: PCOS likely (91.0% confidence). Patterns align with Rotterdam criteria.

DEMO: PCOS Low Probability, Endometriosis Check (Layer 2)

Key Parameter: Chronic Pelvic Pain

Clinical summary

Layer 1 — PCOS probability: 47.0%

Layer 1: PCOS unlikely (47.0%). Checking layer 2...

Layer 2 — Endometriosis probability: 62.0%

Layer 2: Endometriosis possible (62.0% confidence).

DEMO: PCOS & Endometriosis Low Probability (Layer 3)

Clinical summary

Layer 1 — PCOS probability: 47.0%

Layer 1: PCOS unlikely (47.0%). Checking layer 2...

Layer 2 — Endometriosis probability: 32.0%

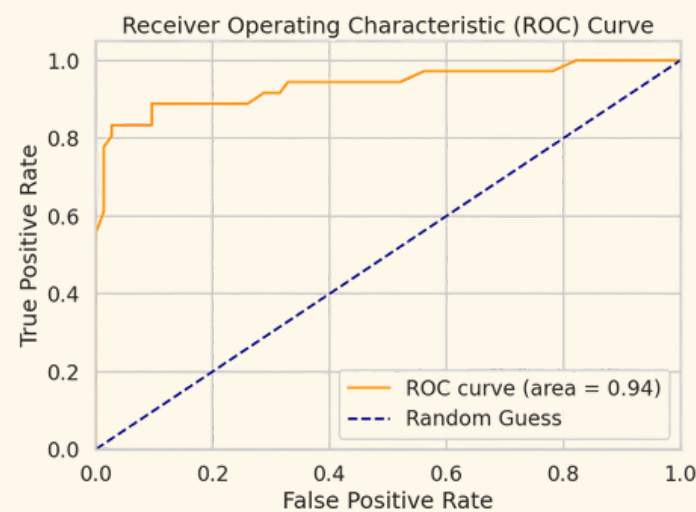
Layer 2: Endometriosis unlikely (32.0%). Layer 3 rule-outs...

TSH and prolactin within typical screening range.

Refer to reproductive endocrinology for comprehensive evaluation.

Impacts and Outcomes

Clinical Validity



AUC of 0.94 on PCOS indicates high sensitivity and specificity
important and expected of medical diagnosis tools

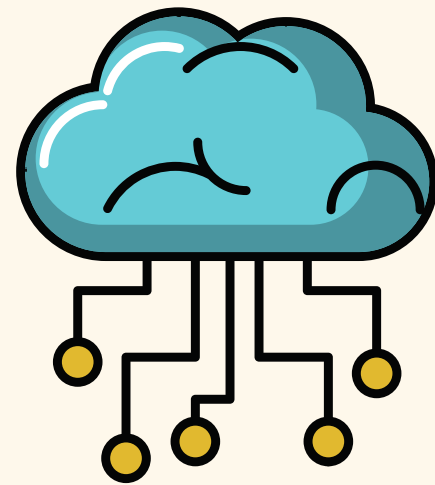
Public Health



Speeds up the timeline of diagnosis
by level of risk based on available
data at each step.

Impacts and Outcomes

Feasibility



Does not depend on cloud services
and can be run natively on any
devices with Python

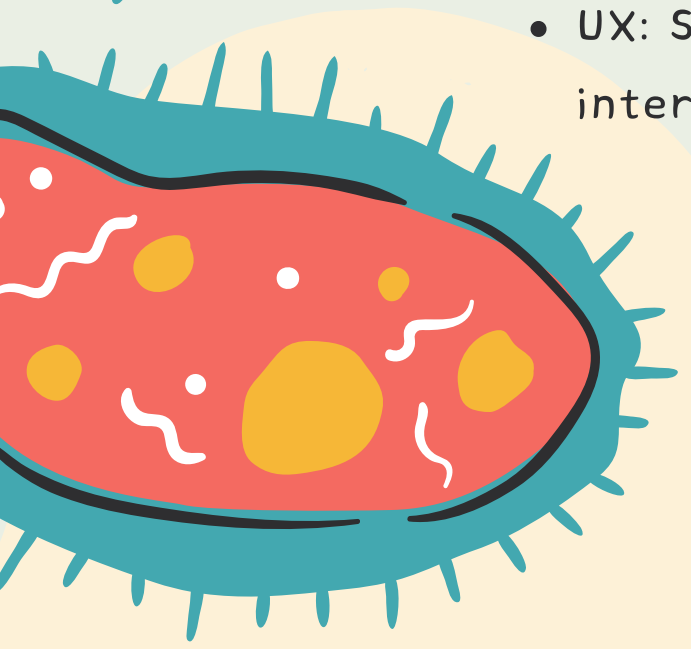
Innovation



Reduces complex variables to only
the most influential and highly
correlated values
Three layered decision tree to
produce a differential diagnosis



Implementation Roadmap



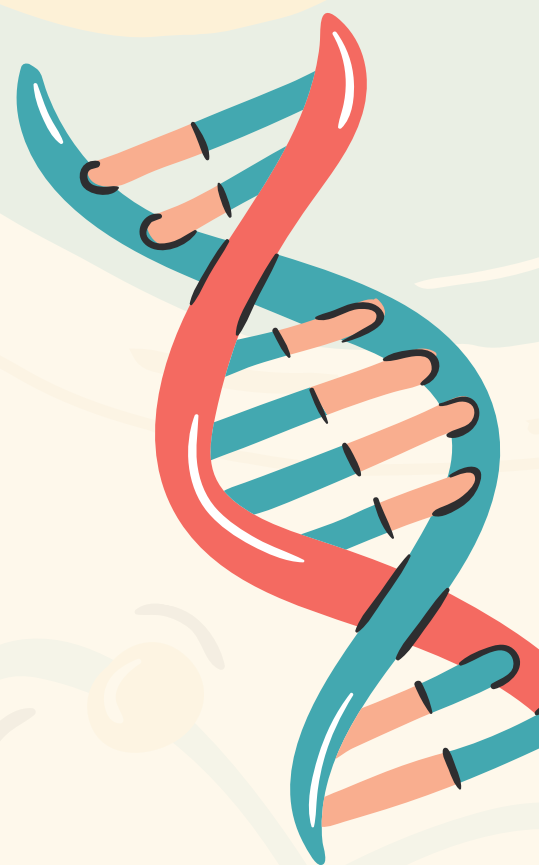
Now (Hackathon)

- Diagnostic Cascade: Sequential 3-layer logic finalized.
- RF Engine: Optimized Random Forest classifiers.
- Guardrails: Out-of-distribution value detection.
- UX: Streamlit-based clinical interface

Short term (0-3 months)

- EHR (Electronic Health Record) Sync: Integration with hospital HL7/FHIR (Fast Healthcare Interoperability Resources) systems.
- Clinical Pilot: Beta testing with 200+ patient cohort.
- Export patient reports as PDF

Mid term (0-3 months)

- Multi-center clinical validation study.
 - Regulation: Pursue SaMD (Medical Device) certification.
 - Diversity: Cross-demographic dataset validation.
 - Platform: Full reproductive health ecosystem.
- 
- 



THANK YOU

Questions?